Biology Class 11 National Book Foundation FBISE Quality Notes
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Chapter 3: Enzymes

Introduction

Metabolism: The sum of all chemical reactions going on in a cell.

Enzymes: Biological polymers (catalysts) that catalyse biochemical reactions. They speed up reactions without being consumed and exist in the cell as colloids.

- Enzymes are made by living cells according to the base sequence in DNA, but they can act outside the cell (in vitro). They are now also made by recombinant DNA technology.

Enzyme structure (3.1)

- All enzymes are **globular proteins** made of one or more polypeptides, linear chains of amino acids folded into a 3D structure.
- **Ribozymes:** the only enzymes made of RNA, found in ribosomes; e.g. peptidyl transferase forms peptide bonds in protein synthesis.

SLO 1: Active site and its components (3.1.1)

- Catalytic activity is located in the **active site**: a specific, charge bearing, three-dimensional cavity.
- Substrate (reactant) attaches to it by non-covalent interactions: hydrogen bonds and hydrophobic interactions.
- Made of only **3-12 amino acids** which may be scattered in the chain but are brought together by secondary and tertiary folding (e.g. aldolase: glycine, histidine, alanine).

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Region of active site	Function
Binding site	Amino acids that make bonds with the substrate and hold it.
Catalytic site	Amino acids that convert substrate into product (catalysis).

- Shape of the active site is designed according to the substrate, so only a particular (sometimes a related) substrate can bind.
- **Holoenzyme:** active enzyme with cofactor attached. **Apoenzyme:** protein part without cofactor; inactive.
- **Pepsin** needs no cofactor: it is secreted as inactive **pepsinogen** with an extra polypeptide fragment blocking the active site; in HCl of stomach the fragment is removed and active pepsin forms.

SLO 2: Types of cofactors (3.1.2)

- A **cofactor** is a non-protein part that helps attach the substrate and takes part in catalysis; final shape of active site forms after it binds.

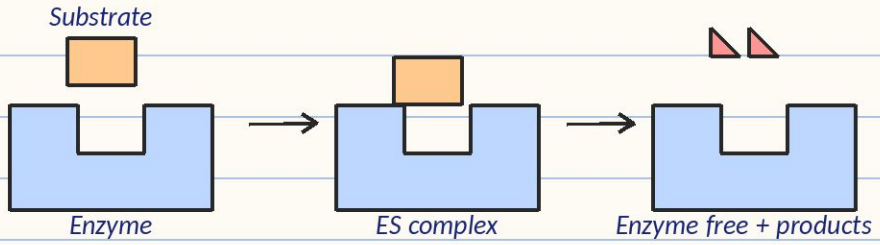
Cofactor	Details and examples
Inorganic ions (activators)	Metal ions Fe^{2+}, Mg^{2+}, Cu^{2+}, Zn^{2+}; detachable, attach when substrate binds. e.g. glucose + ATP with hexokinase + Mg^{2+} gives glucose-6-phosphate + ADP.
Coenzymes (organic)	Derivatives of vitamins; detachable. ATP, NAD^+, FAD^+. e.g. ethyl alcohol + NAD^+ with alcohol dehydrogenase gives acetaldehyde + NADH.
Prosthetic group (organic)	Covalently bonded, permanently attached, does not detach after reaction. e.g. iron containing porphyrin ring in cytochromes.

SLO 3: Mechanism of enzyme action (3.2)

- E + S to ES complex to EP complex to E + P. The enzyme is released unchanged and used again.

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Lock and key (Fischer 1894): rigid active site



Induced fit (Koshland 1959): flexible active site

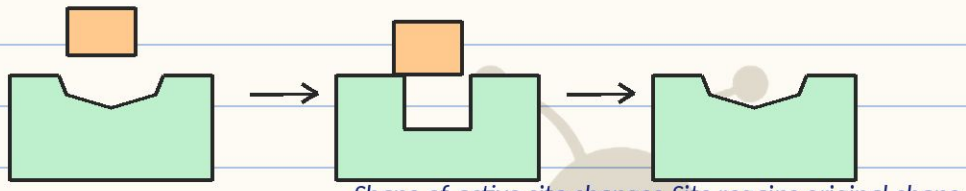


Fig: Models of enzyme action

Lock and key model	***Induced fit model***
Emil Fischer, 1894	*Koshland, 1959*
Enzyme = lock, substrate = key; active site is rigid and complementary to substrate	*Active site is flexible; modified as substrate interacts; regains shape after reaction*
Precisely fits one substrate; explains specificity	*Does not fit completely at first; conformational change improves binding*
Non-regulatory enzymes; few enzymes: sucrase, maltase	*Regulatory / allosteric enzymes: hexokinase*
Absolute specificity (urease: urea + water to ammonia + CO_2)	*Related substrates can bind; one enzyme can do related reactions (carbonic anhydrase: Hb + O_2; CO_2 + H_2O to H_2CO_3)*

SLO 4: Enzyme catalysis with examples

- *Urease: urea + water to ammonia + carbon dioxide (absolute specificity).*
- *Carbonic anhydrase: adds O_2 to haemoglobin and controls formation of carbonic acid and bicarbonate in blood.*
- *Hexokinase: glucose + ATP to glucose-6-phosphate + ADP (Mg^{2+} cofactor). Alcohol dehydrogenase: ethanol to acetaldehyde with NAD^+.*

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SLO 5: Energy of activation (3.2.2)

Energy of activation: *The energy that must be added to cause molecules to react with one another; energy required to break existing bonds and begin the reaction.*

- *In non-living systems heat is used. Living cells cannot use large heat because their proteins are temperature sensitive and about 1000 reactions run at a time; cells work under* ***isothermal*** *conditions.*
- *Enzyme (E) reacts with reactant A to form AE transitional complex, which reacts with B to give AB and free E. The enzyme* ***lowers the activation energy*** *by changing an energy dependent process into an energy independent one.*

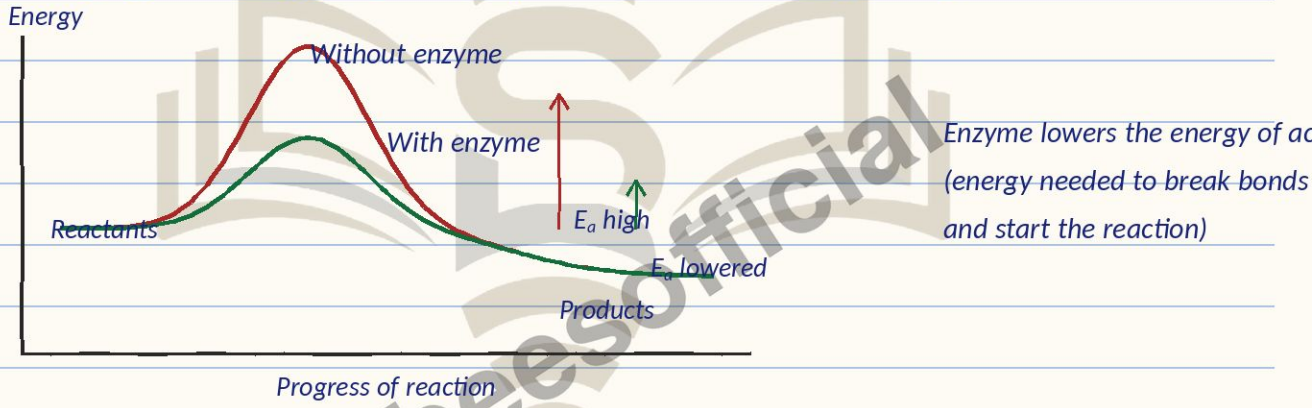


Fig: Energy of activation

SLO 6: Effect of temperature (3.3.1)

- *Rate of reaction = amount of substrate changed or product formed per unit time. Factors: temperature, pH, enzyme concentration, substrate concentration.*
- *Heating increases molecular motion and collisions; a rise of* ***10 C doubles the rate*** *up to a limit.*
- ***Optimum temperature:*** *temperature of maximum activity. Above it enzyme is* ***denatured*** *(secondary and tertiary structure disrupted, active site lost; hydrogen bonds most sensitive); that temperature = maximum temperature.*
- *Human enzymes: optimum 37-38 C. Thermophilic bacteria of hot springs: 70 C or higher (their enzymes are used in biological washing powders).*

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- *Near or below freezing enzymes are **inactivated, not denatured**; they work again when temperature rises (minimum temperature).*

SLO 7: Effect of pH (3.3.2)

- ***Optimum pH:** narrow pH range of maximum rate. pH changes charges on amino acid side chains that maintain tertiary / quaternary structure. Slight change = temporary inactivation (ionization); extreme change = denaturation.*
- *Most enzymes: optimum pH 6-8.*

Enzyme	***Optimum pH / note***
Pepsin	*Acidic, pH 2 (stomach).*
Trypsin	*Alkaline, pH 8; inactive at pH 2.*
Papain (green papaya latex)	*Acts in both acidic and alkaline media; wide pH range 5-8; optimum temperature 65 C; used to protect plants from insects.*

SLO 8: Effect of enzyme concentration (3.3.3)

- *If substrate is unlimited and pH and temperature are constant, rate is **directly proportional** to enzyme concentration (more active sites available). The effect continues until equilibrium state; after that more enzyme gives no further effect.*

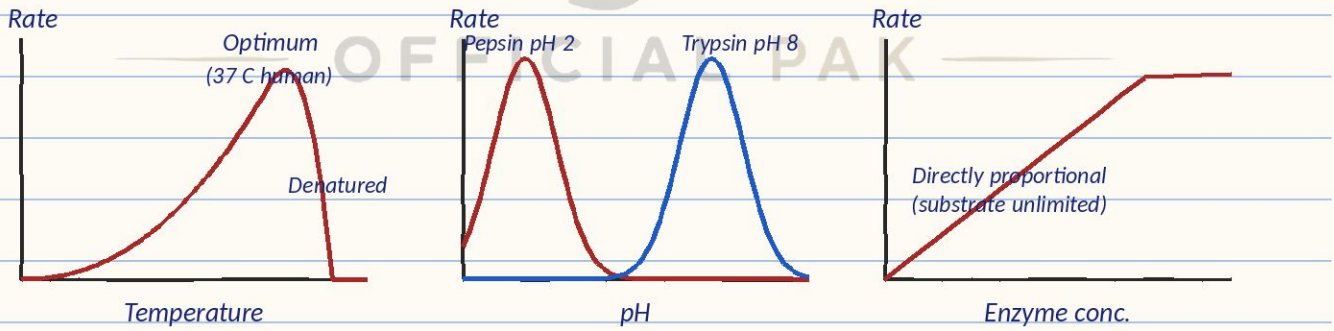


Fig: Effect of temperature, pH and enzyme concentration

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SLO 9, 10: Enzyme inhibition and inhibitors (3.4)

Enzyme inhibition: *The phenomenon in which an enzyme fails to catalyse a reaction. Inhibitors are molecules which react with enzyme but are not converted into the desired products.*

- *Inhibition is a normal part of regulation of enzyme activity in cells, but external inhibition can be dangerous to life.*
- ***Molecules acting as inhibitors:*** *poisons, cyanides, antibodies, anti-metabolites, penicillin, sulpha drugs, heavy metal salts, malonate, nerve gas (DFP).*
- *Significance: regulation of metabolism (feedback); drugs / antibiotics against bacteria (penicillin blocks active site of enzyme unique to bacteria, human enzymes unaffected); poisons and venoms.*

SLO 11: Competitive and non-competitive inhibitors

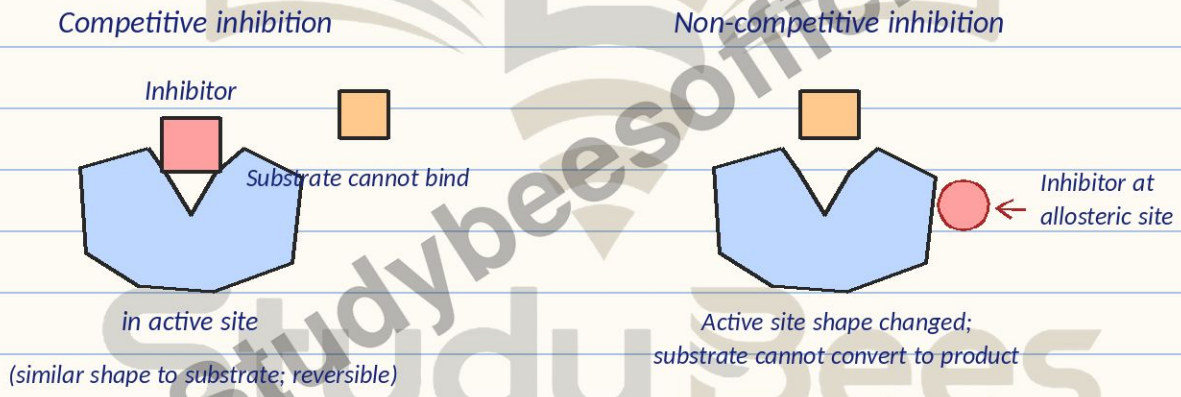


Fig: Competitive and non-competitive inhibition

Competitive inhibition	***Non-competitive inhibition***
Inhibitor competes with substrate for the active site	*Inhibitor binds at another site, the allosteric site*
Inhibitor structurally similar to substrate	*Not similar to substrate*
Reversible; depends on relative concentrations; more substrate reduces inhibition	*Reversible (temporary) or irreversible (denatures enzyme); substrate concentration has no effect*

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Competitive inhibition	Non-competitive inhibition
Example: malonate on succinate dehydrogenase (succinate to fumarate); sulphonamides	Reversible: feedback inhibition. Irreversible: cyanides, salts of heavy metals (Hg^{2+}, Ag^{+}, Cu^{2+})
Supports lock and key hypothesis; used as drugs	Prevent formation of enzyme-product complex

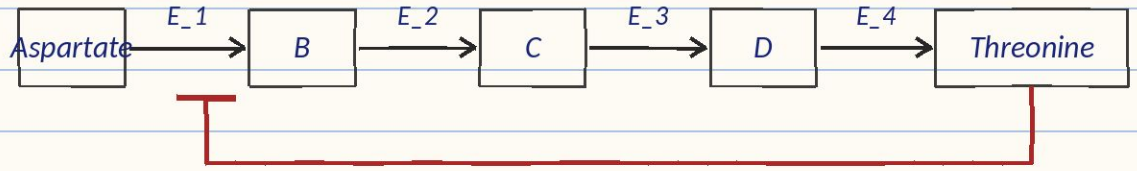
- **Cyanides:** combine with iron of the prosthetic group and inhibit cytochrome oxidase, blocking cellular respiration. **Heavy metal ions:** combine with thiol (-SH) groups and break disulphide bridges, so tertiary structure is lost and the enzyme is denatured.
- Examples (competitive): antibodies, antimetabolites, penicillin, iodoacetate, malonate, CoA (high concentration). Examples (non-competitive): acetaldehyde, DFP (nerve gas), mercury, silver, copper, cyanide.
- **Snake venom** inhibits cholinesterase and cytochrome oxidase. Types: haemotoxic (heart, blood vessels), neurotoxic (nerves, brain), cytotoxic (local action at bite).

SLO 12: Feedback inhibition (3.4.3)

Feedback inhibition: Inhibition of an enzyme by its own product; a type of reversible non-competitive inhibition and part of normal regulation of metabolic pathways.

- Example: aspartate becomes threonine in five enzymatic steps. Excess threonine (end product) binds an allosteric site on enzyme 1 so its active site cannot bind aspartate. When threonine is used up it leaves the site and the pathway resumes.

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Excess end product binds allosteric site of E_1 and inhibits it (-)
Fig: Feedback inhibition

SLO 13: Classification based on reaction type (3.5.1)

- *By International Union of Biochemistry (1961): six groups.*

Group	Reaction and example
1. Oxidoreductases	Oxidation / reduction by adding or removing electrons or H^+. e.g. cytochrome oxidase.
2. Transferases	Transfer a functional group (not hydrogen) from one substrate to another. e.g. hexokinase moves phosphate from ATP to glucose.
3. Hydrolases	Break large molecules by adding water. e.g. pepsin, renin, trypsin, digestive enzymes.
4. Lyases	Break specific covalent bonds, remove groups without hydrolysis. e.g. histidine decarboxylase gives CO_2 + histamine.
5. Isomerases	Intramolecular rearrangement forming one isomer from another. e.g. phosphohexose isomerase: glucose-6-phosphate to fructose-6-phosphate.
6. Ligases (synthetases)	Join two molecules using energy from ATP hydrolysis. e.g. polymerases make DNA / RNA.

- ***Naming of enzymes:*** *add "ase" to substrate (proteases, lipases); by reaction type (oxidases, reductases); by both (DNA polymerase); by substance synthesized (rhodonase).*

SLO 14: Classification based on substrate (3.5.2)

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Group	Examples and action
Proteases	Pepsin, trypsin (large polypeptides to small polypeptides / peptones); amino- and carboxypeptidases (peptones to dipeptides); erepsin (dipeptides to amino acids).
Lipases	Hydrolyse lipids into fatty acids and glycerol; pancreatic lipase.
Carbohydrases	Amylase (starch / glycogen to maltose); cellulase (cellulose to cellobiose); maltase (maltose to glucose); sucrase (sucrose to glucose + fructose); lactase (lactose to galactose + glucose).
Nucleases	RNAase (RNA to ribonucleotides); DNAase (DNA to deoxyribonucleotides); ATPase (hydrolysis of ATP in muscles).

Diagnostic uses of enzymes

Enzyme	Diagnosis of
Aldolase	Muscular dystrophy, viral hepatitis, advanced prostate cancer
Creatine phosphokinase	Damage to muscle cells
Gamma-glutamyl transpeptidase	Liver function
Lactic dehydrogenase	Differentiating heart attack, anaemia, lung injury, liver disease
Lipase	Damage to the pancreas

STEAM activity 3.1: Immobilized enzymes (lactose-free milk)

- Lactase is mixed with sodium alginate and dropped into calcium chloride to make hard beads (immobilized lactase). Beads are put in milk; lactose is broken into galactose and glucose (detected by Benedict's test). Beads are filtered out, so the milk does not contain lactase and the enzyme can be reused; useful for lactose intolerance.

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Chapter 3: Short Questions with Answers (SLO based)

Q1. [SLO 1] What are enzymes? Why are they called biological catalysts?

Ans: Enzymes are biological polymers (proteins) that catalyse biochemical reactions. They speed up reactions without being consumed.

Q2. [SLO 1] What are ribozymes?

Ans: Enzymes made of RNA only, found in ribosomes. Example: peptidyl transferase forms peptide bonds during protein synthesis.

Q3. [SLO 1] What is the structure of an enzyme?

Ans: Enzymes are globular proteins made of one or more polypeptides (linear amino acid chains folded into a 3D structure). Some also have a non-protein cofactor.

Q4. [SLO 1] What is an active site? How many amino acids form it?

Ans: A specific, charge bearing 3D cavity where catalysis occurs; made of 3-12 amino acids brought together by folding.

Q5. [SLO 1] Differentiate between binding site and catalytic site.

Ans: Binding site amino acids bond with the substrate and hold it; catalytic site amino acids convert the substrate into product.

Q6. [SLO 1] Differentiate between apoenzyme and holoenzyme.

Ans: Apoenzyme: protein part without cofactor, inactive. Holoenzyme: active enzyme with cofactor attached.

Q7. [SLO 1] Explain the enzyme pepsin which does not require a cofactor.

Ans: Pepsin is secreted as inactive pepsinogen with an extra polypeptide fragment blocking its active site. In the HCl of the stomach the fragment is removed and active pepsin forms.

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Q8. [SLO 2] What is a cofactor? Name its types.

Ans: A non-protein part needed by some enzymes for substrate attachment and catalysis. Types: inorganic ions, coenzymes and prosthetic groups.

Q9. [SLO 2] What is a prosthetic group? Give an example.

Ans: An organic cofactor covalently and permanently attached to the enzyme. Example: iron containing porphyrin ring in cytochromes.

Q10. [SLO 2] Differentiate between prosthetic group and coenzyme.

Ans: Prosthetic group is covalently bound and permanent. Coenzyme (vitamin derivative, e.g. NAD^+, FAD^+, ATP) is detachable and binds when substrate binds.

Q11. [SLO 2] Differentiate between inorganic and organic cofactors.

Ans: Inorganic: metal ions (Fe^{2+}, Mg^{2+}, Cu^{2+}, Zn^{2+}) called activators. Organic: coenzymes and prosthetic groups.

Q12. [SLO 3] What is the mechanism of enzyme action?

Ans: Substrate binds the active site forming ES complex, converts to product (EP complex), product is released and the enzyme is free to act again.

Q13. [SLO 3] State the lock and key model.

Ans: Fischer (1894): enzyme is a rigid lock and substrate is the key whose shape fits the active site exactly.

Q14. [SLO 3] State the induced fit model.

Ans: Koshland (1959): the flexible active site is moulded to a precise shape when the substrate binds, and returns to original shape afterwards.

Q15. [SLO 3] Differentiate between lock and key model and induced fit model.

Ans: Lock and key: rigid site complementary to substrate; explains specificity. Induced fit: flexible site undergoes a conformational change to improve binding; related substrates can bind.

Q16. [SLO 3] What is absolute specificity? Give an example.

Ans: When an enzyme catalyses only one substrate, e.g. urease (urea + water to ammonia + CO_2).

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Q17. [SLO 3] Name two enzymes that follow lock and key and induced fit models.

Ans: Lock and key: sucrase, maltase (non-regulatory). Induced fit: hexokinase, carbonic anhydrase (regulatory / allosteric).

Q18. [SLO 5] What is energy of activation?

Ans: The energy that must be added to make molecules react; energy needed to break existing bonds and begin the reaction.

Q19. [SLO 5] What is the role of free energy of activation in a chemical reaction?

Ans: Molecules do not react unless activated; this energy starts the reaction. Enzymes lower it so reactions occur at body temperature.

Q20. [SLO 5] Why cannot living cells use heat as energy of activation?

Ans: Cells contain temperature sensitive proteins and run about 1000 reactions at a time; they work in isothermal conditions.

Q21. [SLO 5] How does an enzyme lower the energy of activation?

Ans: It forms a transitional complex (AE) with the reactant, changing an energy dependent process into an energy independent one.

Q22. [SLO 6] List the external conditions which affect the rate of enzyme reaction.

Ans: Temperature, pH, enzyme concentration and substrate concentration.

Q23. [SLO 6] What is optimum temperature? What happens above it?

Ans: Temperature of maximum enzyme activity. Above it the enzyme is denatured: its secondary and tertiary structures and active site are lost.

Q24. [SLO 6] Compare the optimum temperatures of enzymes of humans and thermophilic bacteria.

Ans: Human enzymes: about 37-38 C. Thermophilic bacteria of hot springs: 70 C or higher.

Q25. [SLO 6] What is the effect of low temperature on enzymes?

Ans: Enzymes are inactivated, not denatured, near or below freezing; they regain activity when temperature rises.

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Q26. [SLO 6] By how much does a 10 C rise affect the rate of reaction?

Ans: It doubles the rate, up to a certain limit.

Q27. [SLO 7] Describe the range of pH at which human enzymes function.

Ans: Most enzymes work best between pH 6 and 8; exceptions: pepsin pH 2, trypsin pH 8.

Q28. [SLO 7] Compare the optimum pH of pepsin, trypsin and papain.

Ans: Pepsin: acidic (pH 2). Trypsin: alkaline (pH 8), inactive at pH 2. Papain: acts in both acidic and alkaline media (pH 5-8).

Q29. [SLO 7] How does pH affect enzyme activity?

Ans: pH changes the charges on amino acid side chains maintaining tertiary and quaternary structure; slight change inactivates temporarily, extreme change denatures.

Q30. [SLO 8] How does enzyme concentration affect the rate of reaction?

Ans: With unlimited substrate and constant pH and temperature, rate is directly proportional to enzyme concentration (more active sites), until equilibrium.

Q31. [SLO 9] What are enzyme inhibitors? Name the molecules which act as inhibitors.

Ans: Molecules that react with enzymes but are not converted to desired products. Examples: poisons, cyanides, antibodies, anti-metabolites, penicillin, sulpha drugs, heavy metals.

Q32. [SLO 9] What is the significance of enzyme inhibition?

Ans: It regulates enzyme activity in cells (feedback) and is used in drugs against bacteria; external inhibitors like poisons can be dangerous.

Q33. [SLO 11] What is the importance of competitive enzyme inhibitors?

Ans: They support the lock and key hypothesis, show that substances similar to the substrate are not acted upon, and are used as drugs (sulphonamides) against bacterial pathogens.

Q34. [SLO 11] How can competitive inhibition be reduced?

Ans: By increasing the substrate concentration relative to the inhibitor, since it is reversible.

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Q35. [SLO 11] Differentiate between competitive and non-competitive inhibitors.

Ans: Competitive inhibitor is similar to the substrate and binds the active site; non-competitive binds the allosteric site, changing the enzyme shape.

Q36. [SLO 11] What is an allosteric site?

Ans: A binding site other than the active site where a non-competitive inhibitor attaches.

Q37. [SLO 11] Describe cyanides as irreversible non-competitive inhibitors.

Ans: Cyanides combine with the iron of the prosthetic group and inhibit cytochrome oxidase, blocking cellular respiration, so they are potent poisons.

Q38. [SLO 11] Describe ions of heavy metals as irreversible non-competitive inhibitors.

Ans: Hg^{2+}, Ag^{+}, Cu^{2+} combine with thiol (-SH) groups and break disulphide bridges, so the tertiary structure is lost and the enzyme is denatured.

Q39. [SLO 11] Differentiate between reversible and irreversible non-competitive inhibitors.

Ans: Reversible ones inactivate temporarily (feedback inhibition). Irreversible ones permanently denature the enzyme (cyanides, heavy metals).

Q40. [SLO 12] What is feedback inhibition? Give an example.

Ans: Inhibition of an enzyme by its own end product. Example: threonine inhibits the first enzyme of the aspartate to threonine pathway.

Q41. [SLO 13] Name the six groups of enzymes classified by reaction type.

Ans: Oxidoreductases, transferases, hydrolases, lyases, isomerases and ligases.

Q42. [SLO 13] What are transferases? Give an example.

Ans: Enzymes that transfer a functional group (other than hydrogen) from one substrate to another, e.g. hexokinase transfers phosphate from ATP to glucose.

Q43. [SLO 13] What are isomerases and ligases? Give examples.

Ans: Isomerases rearrange atoms inside a molecule (phosphohexose isomerase). Ligases join two molecules using ATP (polymerases).

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Q44. [SLO 13] How are enzymes named?

Ans: By adding "ase" to substrate (lipase), by reaction type (oxidase), by both (DNA polymerase), or by substance synthesized (rhodonase).

Q45. [SLO 14] Name the enzymes which digest starch, maltose, sucrose, lactose and cellulose.

Ans: Amylase, maltase, sucrase, lactase and cellulase respectively.

Q46. [SLO 14] Give the action of erepsin, nucleases and lipases.

Ans: Erepsin digests dipeptides into amino acids. Nucleases break DNA and RNA into nucleotides. Lipases hydrolyse lipids into fatty acids and glycerol.

Q47. [SLO 14] Give two diagnostic uses of enzymes.

Ans: Creatine phosphokinase shows muscle cell damage; lipase shows pancreas damage (also aldolase, lactic dehydrogenase).

Q48. [Extra] Name three types of snake venom.

Ans: Haemotoxic (heart and cardiovascular system), neurotoxic (nervous system), cytotoxic (local at the bite). Venom inhibits cholinesterase and cytochrome oxidase.

Q49. [Extra] Why are many enzymes secreted in inactive form?

Ans: To protect cell proteins from being digested (e.g. pepsinogen becomes pepsin only in the stomach).

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Chapter 3: Long Questions with Answers (SLO based)

Q1. [SLO 1] Describe the structure of enzyme. Explain the role and component parts of the active site.

Ans:

- *Enzymes are globular proteins of one or more polypeptides folded into a 3D structure (ribozymes are RNA).*
- *Active site: specific, charge bearing cavity of 3-12 amino acids brought together by folding; catalytic activity occurs here; substrate binds by hydrogen bonds and hydrophobic interactions.*

Part	Role
Binding site	Amino acids that bond with and hold the substrate.
Catalytic site	Amino acids that convert substrate to product.

- *Cofactor (if needed) completes the shape of the active site; enzyme with cofactor = holoenzyme, without = apoenzyme (inactive).*

Q2. [SLO 2] Differentiate among the three types of cofactors with examples.

Ans:

Cofactor	Features and example
Inorganic ions	Metal ions Fe^{2+}, Mg^{2+}, Cu^{2+}, Zn^{2+}; detachable activators; Mg^{2+} with hexokinase.
Coenzymes	Organic, vitamin derivatives, detachable; ATP, NAD^{+}, FAD^{+}; NAD^{+} with alcohol dehydrogenase.
Prosthetic group	Organic, covalently and permanently attached; iron porphyrin ring of cytochromes.

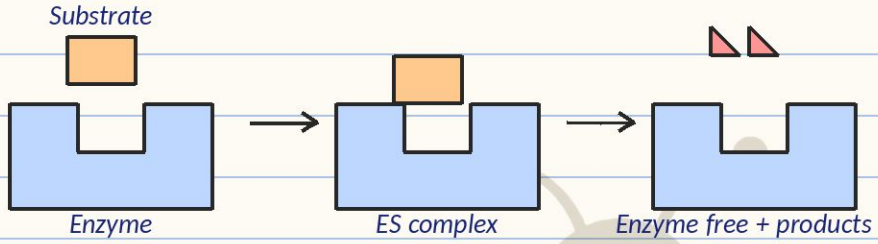
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Q3. [SLO 3] Explain the mechanism of enzyme action through the induced fit model and compare it with lock and key model.

Ans:

- *E + S gives ES complex; S converts to P (EP); P is released and E is free again.*

Lock and key (Fischer 1894): rigid active site



Induced fit (Koshland 1959): flexible active site

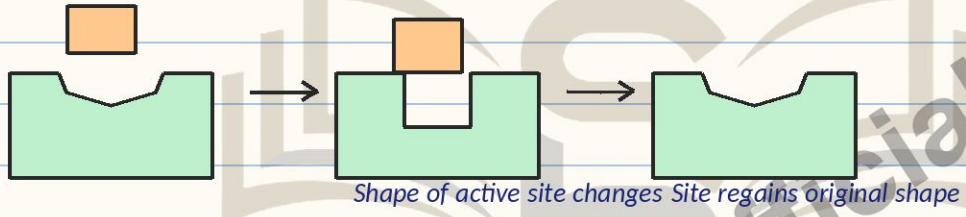


Fig: Models of enzyme action

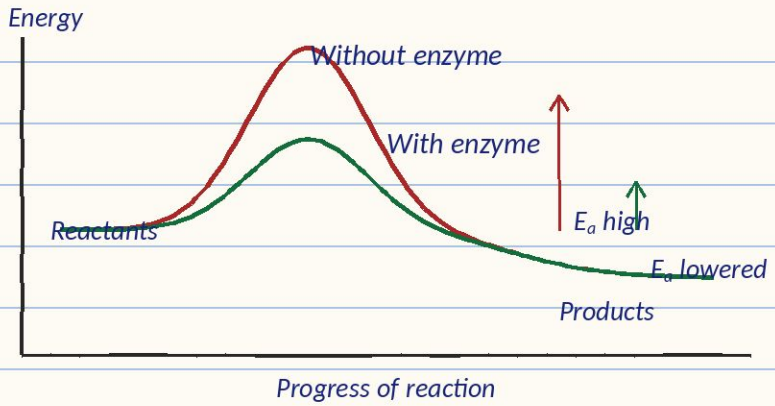
Lock and key	Induced fit
Fischer, 1894	Koshland, 1959
Rigid active site, complement of substrate	Flexible site; moulded by substrate
One specific reaction; absolute specificity (urease)	Related substrates bind; carbonic anhydrase
Non-regulatory: sucrase, maltase	Regulatory / allosteric: hexokinase

Q4. [SLO 4, 5] Define energy of activation and discuss through a graph how an enzyme speeds up a reaction by lowering it.

Ans:

- *Energy of activation = energy needed to break existing bonds and start a reaction. Heat cannot supply it in cells (temperature sensitive proteins, about 1000 reactions, isothermal conditions).*

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Enzyme lowers the energy of ac
(energy needed to break bonds
and start the reaction)

Fig: Energy of activation

- *Enzyme forms a transitional AE complex with reactant A, which then reacts with B to give AB and free E (A + E to AE; AE + B to AB + E). The graph shows a lower peak with enzyme, so more molecules can react in the same time and the reaction speeds up.*

Q5. [SLO 6] Explain the effect of temperature on the rate of enzyme action with the example of human and thermophilic bacteria.

Ans:

- *Heating increases motion and collisions; +10 C doubles the rate up to a limit.*
- *Optimum temperature: maximum activity. Above it enzyme denatures (secondary and tertiary structure lost; hydrogen bonds most sensitive) = maximum temperature.*
- *Human enzymes: optimum 37-38 C. Thermophilic bacteria of hot springs: 70 C or more; their enzymes are used in washing powders for hot washes.*
- *At or below freezing enzymes are only inactivated and recover on warming (minimum temperature).*

Q6. [SLO 7] Investigate the effect of pH on enzyme activity. Compare the optimum pH of trypsin, pepsin and papain.

Ans:

- *Each enzyme works best in a narrow pH range (optimum pH). pH alters charges on side chains which maintain tertiary and quaternary structure. Slight*

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change = temporary inactivation by ionization; extreme = denaturation by breaking ionic bonds.

Enzyme	Optimum pH
Pepsin	pH 2, acidic medium (stomach)
Trypsin	pH 8, alkaline medium; inactive at pH 2
Papain	Acidic and alkaline, pH 5-8; optimum 65 C

- Most enzymes have optimum pH 6-8.

Q7. [SLO 8] Demonstrate that the concentration of enzymes affects the rate of enzyme action.

Ans:

- Keep substrate high (unlimited) and pH and temperature constant.
- One enzyme molecule can convert hundreds of substrates, but slowly. Increasing enzyme concentration makes more active sites available, so conversion becomes faster: rate is directly proportional to enzyme concentration (straight-line graph).
- After equilibrium state (enzyme and substrate concentrations equal) more enzyme has no effect (graph levels off).

Q8. [SLO 9, 10] Describe enzyme inhibition, its types and its significance with examples. Name the molecules which act as inhibitors.

Ans:

- Enzyme inhibition: failure of an enzyme to catalyse a reaction due to inhibitors.
- Inhibitor molecules: poisons, cyanides, antibodies, anti-metabolites, penicillin, sulpha drugs, heavy metals, malonate, DFP.
- Types: competitive and non-competitive (reversible or irreversible); feedback inhibition is reversible non-competitive.
- Significance: normal regulation of metabolism; competitive inhibitors support lock and key and serve as drugs (sulphonamides, penicillin act on bacterial enzymes unique to bacteria); poisons and venoms kill by inhibition (cyanide on

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cytochrome oxidase, snake venom on cholinesterase).

Q9. [SLO 11] Categorize inhibitors into competitive and non-competitive inhibitors with the help of diagrams.

Ans:

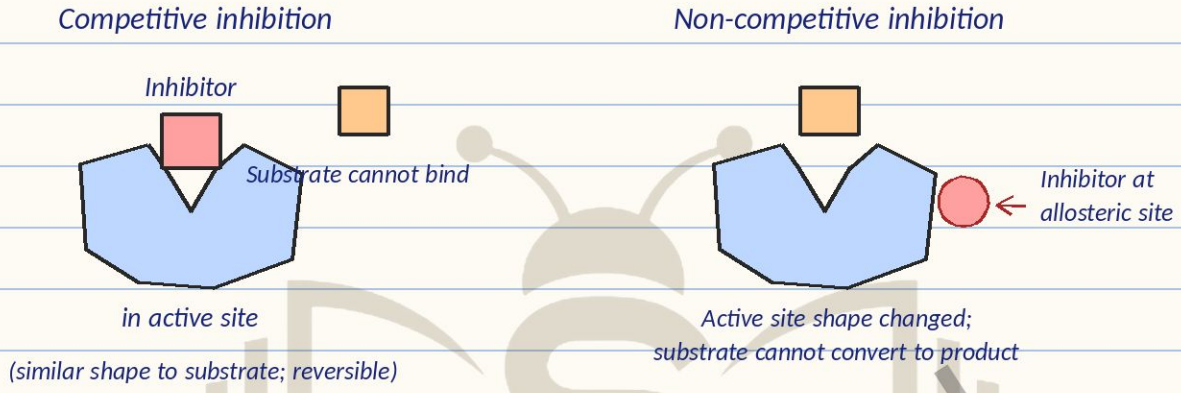


Fig: Competitive and non-competitive inhibition

Competitive	***Non-competitive***
Binds active site; similar to substrate	*Binds allosteric site; not similar*
Reversible; reduced by more substrate	*Reversible or irreversible; substrate has no effect*
Malonate, sulphonamides	*Cyanide, heavy metals, feedback inhibitors*

Q10. [SLO 11] Describe cyanides and ions of heavy metals as irreversible non-competitive inhibitors.

Ans:

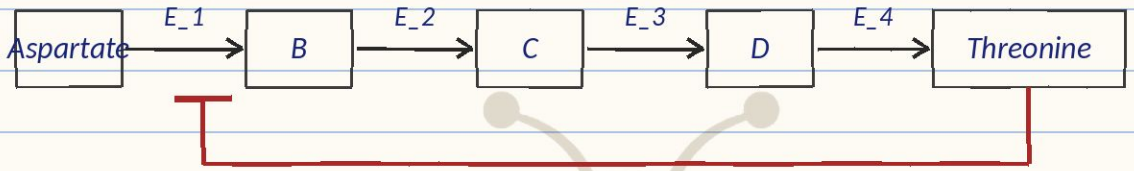
- *Irreversible non-competitive inhibitors destroy the enzyme by altering its shape so substrate cannot bind.*
- ***Cyanides:** potent poisons; combine with iron of prosthetic group of cytochrome oxidase, blocking cellular respiration.*
- ***Heavy metal ions (Hg^{2+}, Ag^{+}, Cu^{2+}):** combine with thiol (-SH) groups, break disulphide bridges necessary for tertiary structure; enzyme is denatured and inactive.*

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Q11. [SLO 12] Explain feedback inhibition with an example and diagram.

Ans:

- *The end product inhibits the first enzyme of its own pathway: reversible non-competitive inhibition; normal regulation of metabolic pathways.*



Excess end product binds allosteric site of E_1 and inhibits it (-)
Fig: Feedback inhibition

- *Aspartate becomes threonine in five enzymatic steps. Excess threonine binds the allosteric site of enzyme 1, so the active site cannot bind aspartate. When threonine is consumed it detaches and the pathway restarts.*

Q12. [SLO 13] Classify enzymes on the basis of reactions catalyzed with examples.

Ans:

Group	Reaction / example
Oxidoreductases	Oxidation-reduction; cytochrome oxidase.
Transferases	Transfer functional group; hexokinase.
Hydrolases	Hydrolysis; pepsin, renin, trypsin.
Lyases	Remove groups without hydrolysis; histidine decarboxylase.
Isomerases	Rearrange atoms; phosphohexose isomerase.
Ligases	Join molecules using ATP; polymerases.

Q13. [SLO 14] Classify enzymes on the basis of the substrates they use (lipases, amylase, proteases etc.).

Ans:

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Group	Examples
Proteases	Pepsin, trypsin, aminopeptidase, carboxypeptidase, erepsin.
Lipases	Pancreatic lipase: lipids to fatty acids + glycerol.
Carbohydrases	Amylase, cellulase, maltase, sucrase, lactase.
Nucleases	RNAase, DNAase, ATPase.

Q14. [Extra] Write the properties and role of enzymes in diagnosis.

Ans:

- Enzymes: protein catalysts, speed up reactions without being used up, specific, work in small amounts, affected by temperature, pH, concentration and inhibitors.
- Diagnosis: aldolase (muscular dystrophy, hepatitis, prostate cancer); creatine phosphokinase (muscle damage); gamma-glutamyl transpeptidase (liver); lactic dehydrogenase (heart attack, anaemia, lung, liver); lipase (pancreas).

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